

Introduction

Artificial Intelligence

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sAlfer Lab
Joint lab on Safety and Security of AI



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Aim of this course

- ▶ Providing basic knowledge of some of the main approaches, methods, and application fields of AI
- ▶ Contributing to turn out graduates capable of designing, developing and operating intelligent systems – *not just users*
- ▶ Course description:
<https://unica.coursecatalogue.cineca.it/...>

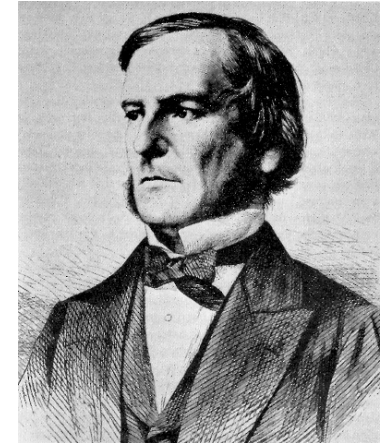
Organisation

- ▶ Website: <https://unica-ai.github.io/>
 - lecture schedule, lecture slides, exercises
- ▶ Mailing list: ai-2026-2027@googlegroups.com
- ▶ Recommended textbook
 - S. Russell, P. Norvig, *Artificial Intelligence – A Modern Approach*, Prentice Hall
 - several editions available at the Faculty Library (also in Italian)
 - <https://aima.cs.berkeley.edu/>
- ▶ Examination
 - **written test**: open questions and exercises on all course topics (20/30)
 - **computer project** on one of the course topics (1 or 2 students, 10/30)

Introduction to Artificial Intelligence

Some of the forefathers

- ▶ G.W. Von Leibniz (1646–1716),
G. Boole (1815–1864):
mechanising logical reasoning
- ▶ A.M. Turing (1912–1954)
 - from the theoretical foundations of programmable computers (*universal Turing machine*)...
 - ...to early speculations on "thinking" machines: *Computing Machinery and Intelligence*, *Mind*, 1950
I propose to consider the question, 'Can machines think?' ...
 - the *Turing test*



Ideas from other disciplines

Several disciplines **studying** different aspects of "intelligence" or intelligent behaviour

- philosophy
- mathematics (logic, computation, probability theory)
- economics
- neuroscience
- psychology
- neurophysiology

Engineering efforts toward **building** machines capable of performing complex tasks

- control theory
- cybernetics

Programmable computers (1940s)

- ▶ Enabled by electronic technology
- ▶ Theoretical foundations: the universal Turing machine
- ▶ **Logical** organisation: the *Von Neumann architecture*
J. Von Neumann (1903–1957)
(inspired by the Turing machine)



The inception of Artificial Intelligence



Dartmouth College Workshop (1956)

We propose that a **2 month, 10 man** study of *artificial intelligence*...

...on the basis of the conjecture that every aspect of **learning** or any other feature of **intelligence** can in principle be so precisely **described** that a **machine** can be made to **simulate** it

An attempt will be made to find how to make machines use **language**, form **abstractions** and **concepts**, solve kinds of **problems** now reserved for **humans**, and **improve themselves**

We think that a **significant advance** can be made in **one or more** of these problems if a carefully selected group of scientists work on it together **for a summer**

What is "intelligence"?

Loosely defined set of capabilities and behaviours

- adapting one's behaviour to novel environmental conditions
- using language
- thinking
- learning
- building abstract models of the world
- planning
- self-consciousness
- ...

What about the term *artificial intelligence*?

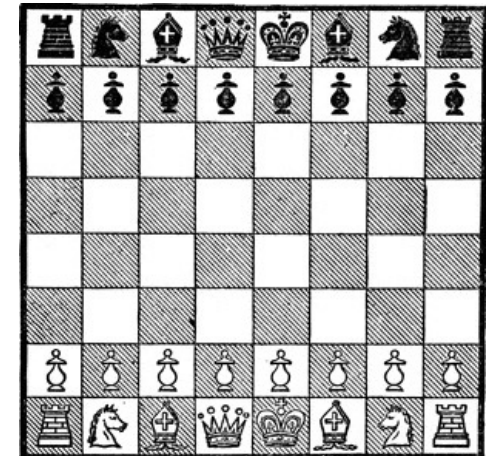
- ▶ "Intelligence": not a rigorously defined concept...
- ▶ "Artificial intelligence"
 - too ambitious?
 - too naive?
 - academic marketing...?
 - a buzzword, nowadays?
 - would a term like "**computational rationality**" have been better?

Philosophical perspectives

- ▶ **Weak AI hypothesis**: machines can **emulate** intelligence (**act** intelligently)
- ▶ **Strong AI hypothesis**: machines can **be** intelligent (whatever this means...)

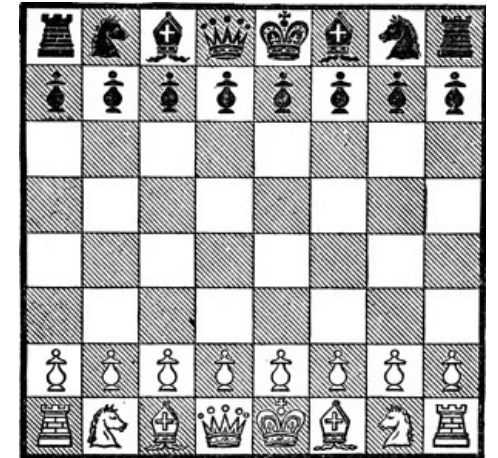
Historical sketch: 1960s – 1980s

- ▶ **Knowledge-driven** approach
 - problem-solving, logical reasoning, search: **abstract symbol processing**
 - applications: theorem proving, game playing (e.g., chess), expert systems (e.g., medical diagnosis), novel programming languages (LISP), etc.
- ▶ Early work on **artificial neural networks**



Failures: the "AI winter" (between the 1960s and 1970s)

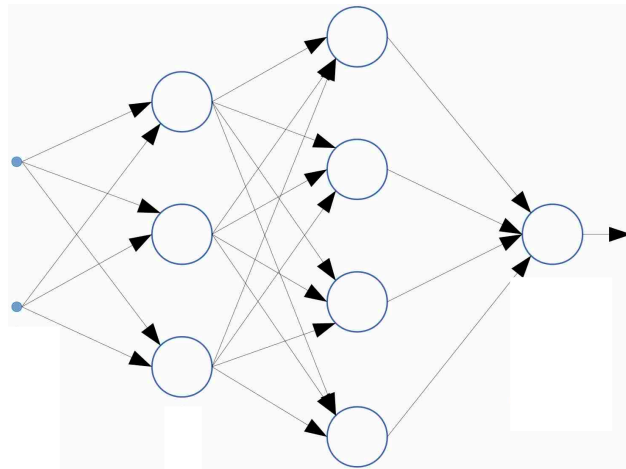
- ▶ Limitations of the knowledge-driven approach
 - "combinatorial explosion" (→ computational complexity)
 - lack of background knowledge
example: natural language processing
*John threw the **ball** to the **window** and broke it*
*John threw the **glass** to the **wall** and broke it*
- ▶ Limitations of early artificial neural networks (1960s)



1980s – present

Connectionist approach

- the return of artificial neural networks (1986: the **backpropagation** algorithm)
- low-level **perception** tasks: *pattern recognition* (signals, sounds, images, etc.)



2000s – present

▶ Data-driven approach

- 2000s: machine learning
- 2010s: deep neural networks (deep learning)
- 2020s: generative AI (generative adversarial networks, transformers, large language models, multimodal models, foundation models, agentic AI)
- General Artificial Intelligence?

▶ Main enabling factors and drivers

- the Internet and Big Data: huge amount of (annotated) data in digital form
- effective statistical machine learning models (CNNs, Transformers, etc.)
- processing power (GPUs, TPUs, data centers)
- economics, geopolitics

Some of AI's current application fields

► Industry

- automation
- robotics
- autonomous vehicles

► Medicine

- diagnostics
- drug design

► Biology

- proteins' secondary structure prediction

► Finance

- automatic (high-frequency) trading

► Military

- autonomous weapons (e.g., UAVs)

► Natural Language Processing

- machine translation
- text generation
- text summarisation

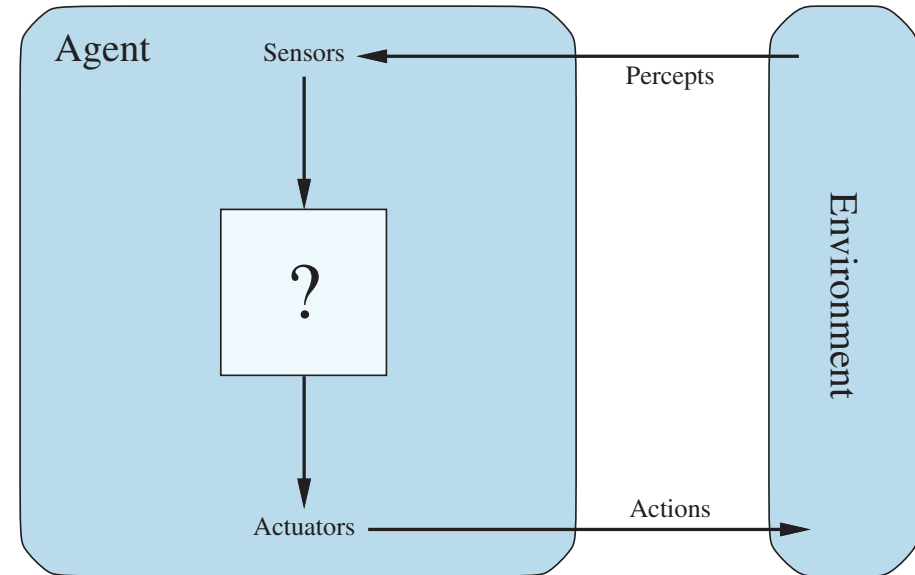
Main approaches to AI so far

- ▶ Human vs rational
- ▶ Thought vs behaviour

	human performance	rationality
thought	systems that think like humans	systems that think rationally
behaviour	systems that act like humans	systems that act rationally ← main approach to AI until recently

The *rational agent* approach

- ▶ **Rationality** depends on
 - **performance measure** (criterion of success)
 - **prior knowledge** of the environment
 - **actions** that can be performed
 - **percept sequence** to date
- ▶ **Rational agent**
 - for each possible percept sequence
 - selecting the action that is **expected** to **maximize** the **performance measure**
 - given the **evidence** (percept sequence) and the **prior knowledge**



Novel issues and risks of AI

Ethical

- lethal autonomous weapons
- ...

Social

- impact on employment
- mass surveillance
- biased decision making
- disinformation
(fake news, deepfakes)
- ...

Legal

- accountability
(e.g., autonomous vehicles)
- ...

Environmental

- energy and water consumption
- ...

Limitations of the rational behaviour approach

- ▶ Achieving perfect rationality (**fully** specified objective)
 - difficult in complex real-world tasks faced by today's AI systems
 - e.g.: chess (really unproblematic...?), self-driving cars
- ▶ **Value alignment** problem
 - machines should pursue **our** objectives, not **theirs**
- ▶ New approach: machines **provably beneficial** to humans?
 - pursuing our objectives, taking into account their **uncertainty**
 - incentive to act cautiously, ask permission, defer to human control

Suggested reading

- ▶ D.R. Hofstadter, *Gödel, Escher, Bach: an eternal golden braid*, Basic Books, 1979
- ▶ S. Zuboff, *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*, Profile Books Ltd, 2019
- ▶ K. Crawford, *The Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*, Yale University Press, 2021
- ▶ N. Cristianini
 - *The Shortcut – Why intelligent machines do not think like us*, CRC Press, 2023
 - subsequent books by the same author

Main disciplines of AI

A broad research field encompassing different approaches and methods (in boldface: the ones covered by this course)

- **heuristic search**
- **knowledge representation**
- **automated reasoning**
- natural language processing
- **machine learning** (just part of AI...)
- computer vision
- robotics

Course syllabus

- ▶ **Problem-solving, knowledge and reasoning** (G. Fumera)
 - Solving problems by searching
 - Knowledge and reasoning: logical languages
 - Knowledge and reasoning under uncertainty: probabilistic models (Bayesian networks)
- ▶ **Introduction to machine learning** (A. Demontis)
 - Classification and regression tasks
 - Decision trees
 - Neural networks
 - Ensemble methods
 - Performance evaluation